

1) Here they asked us to create a "Railway reservation system" and gave us 4 modules. The modules were:

1. Booking
2. Availability checking
3. Cancellation
4. Prepare chart

We were asked to create the modules for representing each data first and to continue with the implementation phase.

2) A matrix game was given with 5 rules. We were asked to implement each of the rules separately.

```

R3 | -  -  -  |
R2 | -  -  -  |
R1 | -  -  -  |
    C1  C2  C3

```

Each of the 9 cells can either be empty or filled with an atom. R3, R2, R1 are the rays that originate from the left. C1, C2, C3 are the rays that originate from the bottom of the box.

Input : Position of the atoms and the rays that gets originated from the outside of the box.

```

Eg.) 3
      3 1
      2 2
      1 3
      3
      R3 C1 C3

```

Output : Print the box.

Rule 1:

A ray that has an atom in its path should print 'H' (Hit) If it does not have any atoms in its path, the ray should pass to the other side.

```

      C1      C3
R3 | -  -  -  | R3
H  | -  X  -  |
R1 | -  -  -  | R1
      C1  H  C3

```

Rule 2 & 3:

A ray that has an atom in its diagonal adjacent position should refract.

```

H  | -  -  -  |
H  | X  -  -  |
R  | -  X  -  |
      R  H  R

```

Input rays: R1, R2, C3

H	-	X	-
R2	-	-	-
	-	-	-

R2 C3

Rule 4:

A ray that has atoms in both of the diagonal adjacent positions should reflect back.

Input ray: C2

	-	-	-
	X	-	X
	-	-	-

R

Input ray: R2

	-	X	-
R	-	-	-
	-	X	-

Rule 5:

The deflection of rays should happen in the order of the input rays.

Input Rays: R3, R2, C1, C3

H	-	X	-
R2	-	-	-
	-	-	-

R2 C3

The final task was to implement these rules for dynamic matrix size.

Input : no of rows, no of columns

Eg.) 4 4 (row & column)

2 (No of atoms)

4 4 (Position of atom)

2 2 (Position of atom)

2 (No of rays)

R4 C2 (Ray number)

H	-	-	-	X
	-	-	-	-
	-	X	-	-
	-	-	-	-

H

The final task was very confusing and it had to handle all the cases. There are chances for a ray to end at the starting position if the number of rows and columns are more than 5.

3) A. Call taxi booking

Input 1:

Customer ID: 1

Pickup Point: A

Drop Point: B

Pickup Time: 9

Output 1:

Taxi can be allotted.

Taxi-1 is allotted

Input 2:

Customer ID: 2

Pickup Point: B

Drop Point: D

Pickup Time: 9

Output 1:

Taxi can be allotted.

Taxi-2 is allotted

(Note: Since Taxi-1 would have completed its journey when second booking is done, so Taxi-2 from nearest point A which is free is allocated)

Input 3:

Customer ID: 3

Pickup Point: B

Drop Point: C

Pickup Time: 12

Output 1:

Taxi can be allotted.

Taxi-1 is allotted

B. Display the Taxi details

Taxi No:	Total Earnings:					
BookingID	CustomerID	From	To	PickupTime	DropTime	Amount

Output:

Taxi-1 Total Earnings: Rs. 400

1	1	A	B	9	10	200
3	3	B	C	12	13	200

Taxi-2 Total Earnings: Rs. 350

2	2	B	D	9	11	350
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These were just sample inputs. It should work for any input that they give.

Those who finished both the modules within 3 hours and if it worked for all the inputs they give, those candidates were given extra modules to work with.

4) Create an engine that can process the user query. The main focus is not the logic but System Design.

1. How the created query engine scales out perfectly even upon adding new features later?

2. How do we create a system that can handle the following inputs and process the input query?

Question:

1. Given a table containing a set of 10 employees with respective fields:

ID	Name	Age	Designation	Department	Reporting To
----	------	-----	-------------	------------	--------------

2. Show all employee data

3. Process the query:

1. Get input from the user until presses exit.

2. Get field value to compare, comparison operator as input

1. If the field value is age (int data type), supported comparators: >, <, !=, ==

2. If the field value is of string data type, supported comparators: 'startswith', 'contains', 'endswith', 'notcontains', 'equals' and 'notequals'.

3. Use 'AND' in default for queries with multiple checks.

4. Eg: age > 30 and age < 50 and department contains finance and reporting to A

4. Show the reporting to hierarchy for the given employee name:

J -> I -> F -> D -> C -> B -> A

5. Show the employees reporting to the given manager.

6. Show summary of Department, Designation, ReportingTo.

5) Question was to build an Employee record management app with 5 tasks to be executed to qualify the round.

we can hard code the employee records in any data structure of our choice.

Columns of the table are: EmpName, Age, Designation, Department, ReportingTo

Task 1: Display all the Employee records in a table format.

Task 2: Search records based on user give criteria.(this task is divided into two sub tasks)

a. display option according to the column type 1.String – Equals, Notequals, Contains, Notcontains, startswith, endswith 2.Integer– <, >, =, !=, between.

b. apply the same for multiple conditions using 'AND' for each condition. (similar to 'where' class in SQL using multiple

columns).

Task 3: print reporting tree of the given employee.

Task 4: print the employees reporting to the given manager.

Task 5: print summary of Department, Designation, ReportingTo.(menu driven)

6) Given an employee data base.

Name, Age, Designation, Department Of ten people.

and Five tasks were given such as

1. Print all employee details.
2. Searching employee details
3. Employees under the given manager name of the department
4. reporting to tree of the given employee name

5,6,7 questions are similar

7) The bank has initially three customers. There were around eight modules .

1. Account Login

2. Purchase

Account Login

Giving customer id and password .Password should be encrypted and stored

Encryption is like A-> B, B->C

a-> b, b->c, c->d

0->1, 1->2

On successful login, it should print the account details

1.Create Gift Card

2. TopUp

3.Transaction History

4.Block

5. Logout

1.Create Gift Card

Gift card with 5 digit card no and 4 digit pin number will be generated

2.TopUp

For topup, amount need to be reduced from main account balance and added to gift card

3.Transaction History

Should print all the transaction details of a particular gift card

4.Block

If the card is blocked, shouldn't be available for topUp,

Purchase.the amount in gift card should be transferred to main account

5.Logging Out

After log out, should go to main module,

2. Purchase

Login to the gift card

Purchase Amount

Then print Available balance

7. Redeem points :

For Every 100 rupee purchase, 1 reward point is added .For 10 reward points, 10 will be added to main account

8. Doing for Multiple gift cards

8) BOMBARDED MAN GAME

Design a system with following functionalities,

1. SET a variable
2. GET a variable
3. UNSET a variable
4. COUNT NUMBERS OF VARIABLE with given value
5. BEGIN – Begins a new transaction
6. ROLLBACK – Roll back all the commands in the open transaction
7. COMMIT – Commit the transaction

EXAMPLE 1:

```
SET a 20
GET a 20
SET b 30
GET b 30
SET a 10
GET a 10
UPDATE c 40 No variable named "c"
SET c 30
COUNT 30 2
COUNT 40 null
UNSET a
```

GET a null

EXAMPLE 2:

GET a null

SET a 30

GET a 30

EXAMPLE 3:

SET a 30

BEGIN

GET a 30

SET a 40

GET a 40

SET b 40

GET b 40

ROLLBACK

GET b null

```

GET a 30
EXAMPLE 4:
BEGIN
SET a 40
SET b 40
SET c 50
COUNT 40 2
BEGIN
COUNT 40 null
COMMIT
COUNT 40 2
BEGIN
SET c 10
GET c 10
ROLLBACK
GET c 50

```

8) In these rounds we were asked to implement the logic for 2 games
– Minesweeper

2. – Breakout a.k.a. Arkanoid a.k.a. Brick-Breaker (you'll find it online)

9) RAILWAY RESERVATION SYSTEM

10) The application was TOLL PAYMENT PROCESSING .

They insisted us to do it in a object oriented language. First they asked the design(what are all the classes and objects & what data structure do you use).

Application description:

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- There are 'n' number of points in a highway out of which some points collect toll.
- Each toll has its own charging scheme according to the vehicles and whether or not they are a VIP user.
- If they are VIP user, 20% discount apply.
- If the vehicle passes 3 toll gates, it has to pay in all the 3 toll gates according to the scheme of respective tolls.

There were 4 modules.

1. Given the details of vehicle type, start and destination.....display the total toll paid during the journey and print the amount after applying the discount.

2. Display the details of all the tolls....like what are all the vehicles(vehicle number) passed that respective toll and the amount each vehicle paid.....and the total amount charged in that toll.

3. Display the details of all the vehicleslike what are all the journeys did it take...the start and destination of the same.....tolls it passed during that journey...amount paid in that journey....and the total amount paid by that vehicle.

4. Assume the highway as a circular path.....we have to find the short route and identify the tolls between that route and calculate the amount.
